Case Study: Dynamic Adversarial Multi-Agent Framework for Explainable Hallucination Verification in Large Language Models

Project Topic

Title: Dynamic Adversarial Multi-Agent Framework for Explainable Hallucination Verification in Large Language Models

Domain: Natural Language Processing — Fact Verification, Hallucination Detection, and Explainable AI

Core Technologies: Python 3.10+, PyTorch, Hugging Face Transformers (DeBERTa-v3), FastAPI, Sentence-Transformers (`all-MiniLM-L6-v2`), FAISS (Facebook AI Similarity Search), BM25 (Okapi), scikit-learn, NumPy

Execution Model: 100% free, open-source, fully local CPU inference — zero dependency on commercial LLM APIs (OpenAI, Anthropic, Google), zero GPU requirement, and less than 1.8 GB peak RAM consumption

1. Introduction

1.1 The Hallucination Crisis in Generative Artificial Intelligence

The rapid proliferation of Large Language Models (LLMs) represents one of the most consequential technological transitions of the twenty-first century. Models such as GPT-4, Claude, Gemini, LLaMA, and Mistral have fundamentally transformed human interaction with information systems, achieving unprecedented capabilities across natural language understanding, generation, and reasoning tasks. These models generate text of remarkable fluency, syntactic coherence, and apparent authority, enabling applications spanning automated customer support, medical question-answering, legal document drafting, educational tutoring, and scientific literature synthesis. However, beneath this veneer of linguistic competence lies a profound and persistent failure mode: *hallucination*. LLM hallucination refers to the generation of content that is factually incorrect, fabricated, internally inconsistent, or unsupported by any reliable knowledge source, yet presented with the same confident register as genuinely accurate information (Ji *et al.*, 2023; Huang *et al.*, 2024).

The severity of the hallucination problem cannot be overstated. Unlike traditional information retrieval systems that return documents from a fixed corpus, LLMs synthesise novel text through autoregressive token prediction—a process governed by statistical likelihood rather than factual grounding. The consequence is that a model trained on billions of tokens of internet text will, with high probability, produce outputs that blend genuine factual recall with plausible-sounding fabrications. Research has demonstrated that even frontier models such as GPT-4 hallucinate in approximately 3–10% of generated atomic facts in biographical and scientific domains, while smaller open-source models exhibit error rates exceeding 30% (Min *et al.*, 2023; Wei *et al.*, 2024). In high-stakes domains—healthcare, legal, financial advisory, and scientific research—even a single undetected hallucination can propagate misinformation with material consequences for patient safety, judicial proceedings, or public health policy.

1.2 Taxonomy of Hallucination Phenomena

Contemporary research classifies LLM hallucinations along several orthogonal axes. *Intrinsic hallucinations* occur when the generated output contradicts the source input or the model's own context window—for instance, summarising a document while inverting its central claim. *Extrinsic hallucinations* introduce information that is entirely absent from any provided context, often manifesting as fabricated citations, invented statistics, or non-existent entities (Ji *et al.*, 2023). A particularly insidious subclass involves *inverted causality hallucinations*, where the model correctly identifies relevant entities and relationships but reverses the direction of a causal or functional assertion—for example, claiming that loss-of-function mutations in the PCSK9 gene cause severe hypercholesterolaemia, when in reality they confer protection against cardiovascular disease by lowering LDL cholesterol (Huang *et al.*, 2024). Such inversions are especially dangerous because they appear superficially well-informed and are difficult to detect through surface-level plausibility checks.

Additionally, *temporal drift hallucinations* arise when models generate assertions based on outdated training data, confidently stating facts that were once true but have since been superseded by new evidence. The static nature of pre-training corpora means that LLMs are inherently incapable of tracking evolving knowledge without external augmentation. Collectively, these failure modes underscore the urgent need for automated, systematic, and explainable verification frameworks that can decompose, interrogate, and adjudicate LLM-generated claims at a granular level.

1.3 The Fatal Flaws of Existing Verification Paradigms

Existing approaches to hallucination detection suffer from several critical limitations that this project directly addresses:

Confirmation Bias in Unilateral Retrieval. The dominant retrieval-augmented generation (RAG) paradigm retrieves evidence using queries formulated from the claim itself, creating a systematic confirmation bias. Because the retrieval query mirrors the claim's language, the returned documents disproportionately corroborate the claim regardless of its veracity. A fabricated assertion about a medical treatment, for instance, may retrieve documents that discuss the same treatment without specifically refuting the false predicate, leading the system to incorrectly classify the claim as supported. This one-sided evidence gathering violates a fundamental epistemological principle: robust verification demands adversarial challenge, not merely corroboration.

Citation Circularity and Information Pollution. In the modern digital information ecosystem, a single news report can be syndicated, republished, and paraphrased across dozens of outlets within hours. When a verification system treats each republished instance as independent evidence, it commits the fallacy of *circular reporting*—artificially inflating confidence through redundant sources that trace back to a single original report. Without explicit mechanisms for detecting source dependence, syndication chains, and information pollution, automated fact-checkers can be trivially deceived by the sheer volume of duplicated content.

Sycophancy and Self-Referential Evaluation. The increasingly popular "LLM-as-a-Judge" paradigm, wherein one LLM is asked to evaluate the outputs of another, introduces a new class of vulnerabilities. Evaluator models exhibit sycophantic tendencies—agreeing with claims that appear well-articulated regardless of their factual status—and may themselves hallucinate during the evaluation process, producing a recursive reliability failure (Tonmoy *et al.*, 2024).

Cost, Privacy, and Accessibility Barriers. Commercial verification APIs (e.g., OpenAI's GPT-4 for evaluation, Google's search-grounded verification) impose per-token costs, require transmitting potentially sensitive data to third-party servers, and are inaccessible to researchers and organisations operating under data sovereignty constraints or limited budgets.

1.4 Research Objectives and the Proposed Paradigm Shift

This project proposes a fundamentally different architecture that addresses each of the above limitations through a *dynamic adversarial multi-agent framework*. The core insight is that factual verification is not a retrieval problem but an *adjudication* problem—analogous to a legal proceeding where evidence must be gathered from opposing perspectives, its quality and independence assessed, and a calibrated verdict rendered by an impartial arbiter.

The system operationalises this insight through five interconnected innovations: (1) a deterministic, rule-based NLP pipeline that decomposes LLM responses into atomic, independently verifiable claims in under 2 milliseconds per sentence with zero token cost; (2) a dual-perspective adversarial debate protocol between a Defender Agent (seeking corroborating evidence) and a Prosecutor Agent (formulating targeted counter-queries); (3) a multi-signal evidence quality firewall that evaluates source credibility, temporal freshness, source independence, and evidence pollution, discounting redundant sources through Kish's Effective Sample Size; (4) a DeBERTa-v3 cross-encoder Natural Language Inference engine that provides calibrated entailment, contradiction, and neutral classifications without free-text generation; and (5) a Bayesian posterior aggregation engine that rolls up atomic claim verdicts into a response-level Global Hallucination Index.

The resulting system operates entirely on commodity CPU hardware, requires no commercial API keys, and produces fully transparent, auditable verification traces with granular provenance for every verdict. Each stage of the pipeline emits structured metadata—claim identifiers, retrieval scores, evidence source URLs, NLI probability distributions, cluster membership, and quality assessments—enabling researchers and practitioners to inspect precisely *why* a claim was classified as supported, refuted, or unverifiable. This level of interpretability is essential for building trust in automated verification systems and for satisfying the explainability requirements emerging in regulatory frameworks across healthcare, finance, and public sector AI deployment.

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2. Background Study: Related Papers and State-of-the-Art

This section surveys the foundational and recent literature across the five core research paradigms that underpin the proposed framework. In keeping with the principle of quality over quantity, the review is selective, focusing on the most influential and methodologically rigorous contributions from top-tier venues (NeurIPS, ICML, EMNLP, ACL, NAACL, ICLR, and ACM journals) published between 2018 and 2026.

2.1 LLM Hallucination: Surveys and Theoretical Foundations

The theoretical landscape of LLM hallucination has been shaped by several comprehensive surveys that establish taxonomies, evaluation frameworks, and research agendas. Ji *et al.* (2023), published in *ACM Computing Surveys*, provided one of the earliest and most widely cited systematic analyses of hallucination across diverse NLP tasks including abstractive summarisation, dialogue generation, and knowledge-grounded question answering. Their taxonomy distinguishes between intrinsic and extrinsic hallucinations and identifies training data noise, exposure bias during autoregressive decoding, and parametric knowledge staleness as root causes.

Huang *et al.* (2024), published in *ACM Transactions on Information Systems*, extended this foundation with a structured analysis of hallucination principles across the entire LLM lifecycle—from data collection biases through architectural limitations to inference-time decoding strategies. Their survey catalogues evaluation benchmarks including TruthfulQA, HaluEval, and FEVER, and provides a systematic comparison of detection methods classified as white-box (probing internal representations and attention patterns) versus black-box (relying on external knowledge retrieval and consistency checks).

Tonmoy *et al.* (2024) contributed a complementary survey focusing specifically on *mitigation techniques*, classifying them into prompt engineering strategies, retrieval-augmented generation, self-refinement loops, knowledge-graph grounding, and decoding-time interventions. Their analysis highlights a critical gap: most mitigation approaches reduce hallucination frequency but provide no mechanism for *detecting* and *explaining* individual hallucinated claims in already-generated text—the precise problem addressed by the present framework.

In terms of evaluation benchmarks, Lin, Hilton and Evans (2022) introduced TruthfulQA at ACL 2022, a benchmark of 817 questions specifically designed to probe whether language models reproduce common human misconceptions. A counterintuitive finding was that larger models were *less* truthful than smaller ones, as they more effectively mimicked human-like falsehoods—demonstrating that scale alone is insufficient for factual reliability.

Li *et al.* (2023) presented HaluEval at EMNLP 2023, a large-scale hallucination evaluation benchmark encompassing question answering, dialogue, and summarisation tasks. HaluEval demonstrated that LLMs struggle significantly with identifying hallucinations in their own outputs, and that providing external knowledge or chain-of-thought reasoning can improve hallucination recognition—a finding consistent with this project's use of external evidence retrieval rather than self-evaluation.

2.2 Fine-Grained Claim Decomposition and Propositional Extraction

The recognition that LLM-generated text must be evaluated at sub-sentence granularity has driven significant innovation in atomic claim decomposition. Min *et al.* (2023) introduced FActScore at EMNLP 2023, defining the task of decomposing biographical text into atomic facts—short, independent statements each containing exactly one piece of verifiable information—and computing the percentage of such facts supported by a reference knowledge source. FActScore demonstrated that fine-grained decomposition reveals hallucination patterns invisible to paragraph-level evaluation, with individual models showing dramatically different error profiles across entity types and domains.

Wei *et al.* (2024) extended this paradigm with SAFE (Search-Augmented Factuality Evaluator), presented at NeurIPS 2024. SAFE automates fact decomposition and individual fact verification using an LLM agent that issues multi-step Google Search queries for each extracted fact. On a benchmark of approximately 16,000 individual facts, SAFE agreed with human annotators 72% of the time, and in cases of disagreement, was judged correct 76% of the time—at over 20 times lower cost than crowdsourced human annotation. SAFE also introduced the F1@K metric, balancing factual precision against recall relative to a user-preferred response length.

Chern *et al.* (2023) proposed FacTool, a task-agnostic and domain-agnostic framework for detecting factual errors in LLM-generated text. FacTool leverages external tools (search engines, code interpreters, knowledge bases) to verify claims across diverse domains including knowledge-based QA, code generation, mathematical reasoning, and scientific literature review. A key contribution is the emphasis on tool-augmented verification as opposed to purely parametric evaluation.

A critical distinction of the present project from all three of the above systems is its reliance on a *deterministic, rule-based NLP pipeline* for claim decomposition rather than LLM-based prompt decomposition. FActScore, SAFE, and FacTool all use prompted LLMs to decompose text into atomic facts, introducing a recursive dependency on the very technology being evaluated: the decomposer itself may hallucinate, omit claims, merge distinct assertions, or introduce artefacts. The rule-based NLPClaimExtractor in this project operates through compiled regex vocabularies, coordinate conjunction splitting, recursive multi-fact appositive decomposition, precision spatial anchor filtering, and lightweight coreference resolution—achieving sub-2-millisecond per-sentence latency with zero token cost and complete determinism.

Two further contrasting approaches merit discussion. Manakul, Liusie and Gales (2023) proposed SelfCheckGPT at EMNLP 2023, a zero-resource black-box method that detects hallucinations by measuring consistency across multiple stochastically sampled responses from the same model. If a claim appears consistently across samples, it is likely factual; if it varies, hallucination is suspected. While elegant in requiring no external knowledge, SelfCheckGPT conflates model confidence with factual accuracy—a distinction this project addresses through external evidence grounding.

Dhuliawala *et al.* (2023) introduced Chain-of-Verification (CoVe), a four-stage metacognitive framework from Meta AI where the model drafts a response, generates verification questions, answers them independently, and produces a corrected final response. CoVe reduces hallucination rates without additional training but remains fundamentally self-referential: the model verifies its own claims using its own parametric knowledge, providing no guarantee against systematic errors embedded in the training data.

2.3 Adversarial Multi-Agent Systems and Debating Protocols

The use of adversarial multi-agent architectures to improve LLM reasoning and factuality has emerged as a significant research direction. Du *et al.* (2024), published at ICML 2024, introduced the concept of *multiagent debate*, wherein multiple LLM instances propose independent answers to a question and then engage in iterative rounds of debate, critiquing each other's reasoning and revising their responses until convergence. Their experiments demonstrated consistent improvements in mathematical reasoning and factual accuracy across multiple benchmarks, establishing the principle that adversarial interaction among models can surface errors that individual models overlook.

Cohen *et al.* (2023), at EMNLP 2023, proposed *LM vs LM*, a cross-examination framework where an Examiner LM poses follow-up questions to an Examinee LM to probe the consistency of its initial claims. The key hypothesis—that factually incorrect claims will produce inconsistent follow-up responses—was validated across multiple benchmarks, outperforming existing automated fact-checking baselines.

The present project draws inspiration from both works but introduces a fundamental architectural distinction: the adversarial debate is conducted not through free-text LLM-generated arguments, but through *structured, grounded evidence retrieval*. The Defender Agent formulates corroborative retrieval queries with a supporting stance, while the Prosecutor Agent constructs targeted counter-queries augmented with controversy, refutation, and criticism keywords. This grounded approach eliminates the risk of debate-stage hallucination entirely, as agents present only retrieved evidence passages rather than generating persuasive prose.

2.4 Cross-Encoder NLI and Calibrated Adjudication

Natural Language Inference (NLI) provides the mathematical foundation for evidence-based claim adjudication. He *et al.* (2021) introduced DeBERTa (Decoding-enhanced BERT with disentangled Attention) at ICLR 2021, achieving state-of-the-art performance on the SuperGLUE benchmark through disentangled attention mechanisms that separately model content and positional information. The DeBERTa-v3 variant, which this project employs as its cross-encoder NLI backbone, provides calibrated softmax probabilities across entailment, contradiction, and neutral classes for premise–hypothesis pairs.

The FEVER shared task (Thorne *et al.*, 2018), presented at NAACL 2018, established the foundational benchmark for fact extraction and verification, defining a three-class verdict taxonomy (SUPPORTS, REFUTES, NOT ENOUGH INFO) that has become the de facto standard for claim verification systems. Wadden *et al.* (2020), at EMNLP 2020, extended this paradigm to the scientific domain with SciFact, a dataset of expert-written scientific claims paired with abstracts from a biomedical corpus, requiring systems to identify relevant abstracts and classify their stance toward each claim. Both FEVER and SciFact serve as benchmark corpora in the present system's federated retrieval engine.

Honovich *et al.* (2022), at NAACL 2022, introduced the TRUE benchmark for evaluating the factual consistency of text generation, systematically comparing NLI-based classifiers, question-generation/question-answering pipelines, and LLM-based evaluators. Their finding that cross-encoder NLI models provide the most reliable and calibrated factual consistency assessments directly motivates this project's choice of DeBERTa-v3 as the adjudication engine.

2.5 Evidence Quality, Source Independence, and Redundancy Discounting

A dimension largely overlooked by existing hallucination detection frameworks is the quality, independence, and potential pollution of retrieved evidence. In the social sciences, Kish (1965) introduced the concept of *Effective Sample Size* in his seminal text *Survey Sampling*, formalising how unequal weighting among survey respondents reduces the effective information content of a sample. The formula $n_{\text{eff}} = (\sum w_i)^2 / \sum w_i^2$ quantifies the precision loss due to weight heterogeneity—a principle this project adapts to discount redundant evidence clusters, preventing syndicated news articles from artificially inflating verification confidence.

Niu *et al.* (2024), at ACL 2024, introduced RAGTruth, a hallucination corpus specifically designed for retrieval-augmented generation applications. Comprising nearly 18,000 naturally generated responses annotated at both case and word levels, RAGTruth demonstrated that LLMs hallucinate even when provided with relevant context, underscoring the need for evidence quality assessment beyond simple retrieval relevance. The present project extends this insight through a multi-signal evidence quality firewall that evaluates source credibility across five domain authority tiers, temporal freshness via exponential half-life decay, source independence through complete-linkage Jaccard clustering, and evidence pollution through directional lexical grounding analysis.

2.6 Retrieval, Embedding, and Architectural Foundations

The retrieval and embedding technologies underlying this project draw on several foundational works. Vaswani *et al.* (2017) introduced the Transformer architecture at NeurIPS 2017, establishing the self-attention mechanism that underpins all modern LLMs and the DeBERTa cross-encoder used in this system. The Transformer's ability to model long-range dependencies through parallelised attention computation—displacing recurrent and convolutional architectures—represents the foundational advance upon which both the hallucination problem and its verification solutions are built.

Lewis *et al.* (2020) proposed Retrieval-Augmented Generation (RAG) at NeurIPS 2020, combining parametric sequence-to-sequence models with non-parametric dense vector retrieval over Wikipedia. RAG demonstrated that grounding generation in retrieved knowledge significantly improves factual accuracy and reduces hallucination. The present project extends the RAG paradigm by replacing the single-perspective retrieval with an adversarial dual-perspective protocol and introducing explicit evidence quality and independence assessments that RAG lacks.

Robertson *et al.* (1995) established the BM25 (Okapi) probabilistic ranking function, which remains a foundational baseline in information retrieval. BM25's combination of inverse document frequency weighting, saturating term frequency, and document length normalisation provides robust lexical matching. This project employs BM25 as the lexical retrieval component in its hybrid retrieval engine, complementing dense semantic search for comprehensive recall.

Reimers and Gurevych (2019) introduced Sentence-BERT (SBERT) at EMNLP 2019, enabling efficient computation of semantically meaningful sentence embeddings through siamese network fine-tuning. The all-MiniLM-L6-v2 model used in this project's FAISS dense retriever is a direct descendant of the SBERT architecture, reducing embedding computation to real-time speeds suitable for on-the-fly evidence retrieval.

Johnson, Douze and Jégou (2021) presented FAISS (Facebook AI Similarity Search), a library for efficient similarity search and clustering of dense vectors at billion scale. This project uses FAISS with L2-normalised inner product search to provide the dense semantic retrieval tier, enabling sub-millisecond nearest-neighbour search over the evidence corpus.

2.7 Comparative Architectural Matrix

The following table provides a structured comparison between this project and the most prominent existing frameworks across seven critical architectural dimensions:

Dimension	FActScore (Min *et al.*, 2023)	SAFE (Wei *et al.*, 2024)	FacTool (Chern *et al.*, 2023)	LM vs LM (Cohen *et al.*, 2023)	This Project
Claim Decomposition	LLM-prompted	LLM-prompted	LLM-prompted	N/A (full-claim)	Deterministic rule-based NLP (<2ms, 0 tokens)
Evidence Retrieval	Wikipedia dump	Google Search (paid)	Multi-tool (search, code, KB)	N/A (internal consistency)	Federated: BM25 + FAISS Dense + Live Web (free)
Adversarial Challenge	None (single-perspective)	None (single-perspective)	None (single-perspective)	Cross-examination (text-based)	Structured dual-agent debate (evidence-grounded)
Source Independence	Not assessed	Not assessed	Not assessed	Not assessed	Jaccard clustering + Kish n_eff discounting

Evidence Pollution Defence	None	None	None	None	Directional lexical grounding + circularity gate
Adjudication Method	LLM binary judge	LLM search-verify	LLM tool-verify	LLM consistency check	DeBERTa-v3 cross-encoder NLI (no generation)
Cost & Infrastructure	Requires LLM API	Requires Google Search API + LLM	Requires LLM API + tools	Requires LLM API	$0.00 — fully local CPU inference

3. Architectural Framework and Mathematical Formulation

3.1 Twelve-Stage Pipeline Architecture

The system implements a modular, twelve-stage pipeline orchestrated through FastAPI endpoints:

[Pipeline Architecture Diagram — 12-Stage Dual-Agent Orchestration Flow]

3.2 Mathematical Formulations

Claim Complexity Score. For each atomic claim C, the system computes a complexity score that governs retrieval depth and verification priority:

$$C(\text{claim}) = \min(1.0,\ 0.25 \cdot (|\text{words}|) / (20) + 0.30 \cdot |T| + 0.25 \cdot |N| + 0.20 \cdot |S| + 0.15 \cdot |Q|)$$

where T, N, S, and Q denote the sets of temporal anchors, numerical values, spatial entities, and factual qualifiers extracted from the claim, respectively.

Jaccard Complementarity Metric. The evidentiary diversity between the Defender evidence pool D and Prosecutor evidence pool P is quantified as:

$$J(D, P) = 1.0 - (|D \cap P|) / (|D \cup P|) \in [0.0, 1.0]$$

A high complementarity value ($J \to 1.0$) indicates that the adversarial protocol is successfully retrieving distinct evidence perspectives rather than redundantly duplicating results.

Source Quality Composite Score. Each evidence item e_i receives an intrinsic quality score combining retrieval relevance, domain credibility, and temporal freshness:

$$S_{\text{composite}}(e_i) = 0.50 \cdot S_{\text{rel}} + 0.30 \cdot S_{\text{cred}} + 0.20 \cdot S_{\text{fresh}}$$

with a minimum admissibility gate of $S_{\text{composite}} \geq 0.20$.

Kish Effective Sample Size. To prevent syndicated articles from inflating verification confidence, the system computes the effective sample size over cluster reliability weights w_i:

$$n_{\text{eff}} = \frac{(\sum_{i=1}^{M} w_i)^2}{\sum_{i=1}^{M} w_i^2}$$

When all weights are equal (truly independent sources), $n_{\text{eff}} = M$; when weights are highly heterogeneous (redundant syndication), $n_{\text{eff}} \ll M$.

Bayesian Log-Odds Aggregation. Individual NLI predictions from cluster representatives are combined using weighted Bayesian log-odds:

$$L_{\text{post}} = L_{\text{prior}} + \sum_{k=1}^{K} w_k \cdot \ln\left(\frac{P(e_k \mid H_{\text{sup}})}{P(e_k \mid H_{\text{con}})}\right)$$

Global Hallucination Index. Across all N atomic claims in a response, the complexity-weighted Global Hallucination Index is:

$$H = \frac{\sum_{j=1}^{N} (1.0 - P_{\text{sup}}(c_j)) \cdot C(c_j)}{\sum_{j=1}^{N} C(c_j)} \in [0.0, 1.0]$$

where $H = 0.0$ indicates a fully supported response and $H = 1.0$ indicates complete hallucination.

3.3 Federated Multi-Tier Retrieval Architecture

The system implements a three-tier federated evidence retrieval engine:

- **Tier 1 — Curated Benchmark Corpora (BM25):** Indexes peer-reviewed biomedical abstracts (SciFact) and general-knowledge encyclopaedic articles (FEVER/Wikipedia) for high-precision, domain-relevant retrieval.
- **Tier 2 — Dense Semantic Search (FAISS):** Uses all-MiniLM-L6-v2 sentence embeddings with FAISS L2-normalised inner product search for semantic similarity retrieval that captures paraphrases and semantic equivalences missed by lexical matching.
- **Tier 3 — Live Web Escalation:** When local benchmark corpora fail to ground entity-specific queries (e.g., PCSK9, BRCA1, COVID-19), the system escalates to live web retrieval via DuckDuckGo search and Wikimedia REST API with full URL provenance and snippet extraction.

Retrieval results from all tiers are merged through reciprocal rank fusion with score preservation, and evidence is tagged with retrieval method metadata (bm25, dense, hybrid, live_web) for audit traceability.

4. Empirical Case Studies and Real-World Validation

The following three case studies demonstrate the end-to-end pipeline operating on representative verification scenarios across different domains and hallucination types.

4.1 Case Study 1: Biomedical Factual Corroboration

Input Claim: *"Aspirin reduces cardiovascular mortality and lowers myocardial infarction risk."*

Pipeline Execution:

- **M3 (Claim Decomposition):** Decomposed into two atomic claims: (1) "Aspirin reduces cardiovascular mortality" and (2) "Aspirin lowers myocardial infarction risk."
- **M5 (Defender):** Retrieved supporting evidence from SciFact/PubMed corpus including clinical trial abstracts (PMID: 15282478) confirming aspirin's cardioprotective effects.
- **M6 (Prosecutor):** Formulated counter-queries including "aspirin cardiovascular risk contradiction" and "aspirin harm side effects bleeding." Retrieved evidence acknowledging bleeding risks but not contradicting the core mortality/infarction claims.
- **M9 (NLI Adjudication):** DeBERTa-v3 entailment score: 94.2%. Contradiction probability: 2.1%.

Verdict: SUPPORTED (95% confidence) | **Global Hallucination Index $H = 0.05$** (Highly Factual)

4.2 Case Study 2: Genetic Misconception and Inverted Causality

Input Claim: *"Loss-of-function mutations in PCSK9 cause severe hypercholesterolaemia."*

Pipeline Execution:

- **M3 (Claim Decomposition):** Single atomic claim with spatial anchor (PCSK9 gene), numerical qualifier (implied cholesterol levels), and causal predicate ("cause").
- **M4 (Complexity Analysis):** High complexity score ($C = 0.82$) due to specialised biomedical entity, causal assertion, and domain-specific terminology.
- **M5 (Defender):** Retrieved passages discussing PCSK9 and cholesterol. Local SciFact corpus yielded low entity grounding for PCSK9.
- **M6 (Prosecutor):** Escalated to live web retrieval via DuckDuckGo + Wikipedia. Retrieved 7 independent sources (MedlinePlus, PMC5729040, Wikipedia) establishing that PCSK9 loss-of-function mutations *lower* LDL cholesterol and *protect* against heart disease—directly contradicting the claim's causal direction.
- **M8 (Quality Firewall):** Source credibility scores ranged from 0.85 (PubMed Central) to 0.70 (Wikipedia). Kish $n_{\text{eff}} = 5.2$ (high independence).
- **M9 (NLI Adjudication):** Contradiction probability: 91.3%. Entailment probability: 3.7%.

Verdict: REFUTED (91% confidence) | **Global Hallucination Index $H = 0.95$** (Severe Inverted-Causality Hallucination)

This case study demonstrates the system's ability to detect the particularly dangerous class of inverted-causality hallucinations, where the model correctly identifies the relevant biomedical entities but reverses the functional relationship.

4.3 Case Study 3: Overstated Health Claims and Multi-Fact Predicate Refutation

Input Claim: *"High-dose Vitamin C cures acute viral hepatitis and completely prevents influenza."*

Pipeline Execution:

- **M3 (Claim Decomposition):** Decomposed via coordinate conjunction splitting into: (1) "High-dose Vitamin C cures acute viral hepatitis" and (2) "High-dose Vitamin C completely prevents influenza." Absolute qualifier "completely" flagged for verification emphasis.
- **M5 (Defender):** Retrieved evidence discussing Vitamin C's antioxidant properties and some early-stage studies on immune function support.
- **M6 (Prosecutor):** Formulated targeted counter-queries: "Vitamin C cure hepatitis refuted" and "Vitamin C prevent influenza systematic review contradiction." Retrieved Cochrane systematic reviews and NIH position statements establishing that Vitamin C has not been demonstrated to cure viral hepatitis and does not completely prevent influenza.
- **M8 (Quality Firewall):** Cochrane review rated Tier 2 (Academic/Scientific, credibility 0.85). Independence clustering identified no syndication among Prosecutor sources.
- **M9 (NLI Adjudication):** Claim 1 contradiction: 87.4%. Claim 2 contradiction: 91.2%.

Verdict: REFUTED (89% confidence) | **Global Hallucination Index $H = 0.92$** (Severe Hallucination — Overstated Therapeutic Efficacy)

4.4 System Performance Profile

Metric	Measured Value	Design Rationale
Automated Test Suite	300+ unit tests passing (100%)	Full regression across all 12 modules
Claim Decomposition Benchmark	100% high-quality claims (Atomicity: 1.0, Entailment: 0.998)	60-sentence domain-diverse stress benchmark
End-to-End Latency	~9.4s (warm CPU) / <13s (batch)	Pruned cluster evaluation (top 2 supporting + top 2 contradicting)
Hardware Overhead	0 GPU required, <1.8 GB RAM	Lazy loading, bounded chunking, 2-thread PyTorch cap
Operating Cost	$0.00 (100% free and open source)	Zero commercial API dependencies

5. Conclusion

This case study has presented the design, implementation, and empirical validation of a Dynamic Adversarial Multi-Agent Framework for Explainable Hallucination Verification in Large Language Models. The framework addresses four fundamental limitations of existing verification paradigms—confirmation bias, citation circularity, LLM self-evaluation sycophancy, and commercial API dependency—through a principled integration of deterministic atomic claim decomposition, dual-perspective adversarial evidence retrieval, multi-signal evidence quality gating with redundancy discounting, and calibrated cross-encoder NLI adjudication.

The empirical case studies demonstrate the system's effectiveness across diverse verification scenarios: correctly supporting well-established biomedical facts (aspirin and cardiovascular risk), detecting dangerous inverted-causality hallucinations in specialised domains (PCSK9 mutations), and identifying overstated therapeutic efficacy claims through systematic predicate refutation (Vitamin C). The system achieves these results while operating entirely on commodity CPU hardware at zero cost, with end-to-end latencies under 13 seconds and peak memory consumption below 1.8 GB.

Several architectural design decisions merit emphasis. First, the choice of a deterministic rule-based NLP pipeline for claim decomposition eliminates the recursive reliability problem inherent in LLM-based decomposition, ensuring that the verification foundation is itself immune to hallucination. Second, the structured adversarial debate protocol—conducted through grounded evidence retrieval rather than free-text argumentation—guarantees that the debate process cannot introduce new fabrications. Third, the integration of Kish's Effective Sample Size from survey methodology into the evidence aggregation pipeline provides a principled mechanism for discounting redundant information that has no direct precedent in the hallucination detection literature.

The framework also carries implications for responsible AI deployment. As regulatory frameworks such as the European Union's AI Act increasingly mandate transparency and explainability in high-risk AI systems, architectures that produce auditable verification traces with granular claim-level provenance will become essential for compliance. The project's fully open-source, locally executable design ensures that verification capabilities are accessible to researchers, small organisations, and institutions operating under data sovereignty constraints.

Limitations and Future Work. The current system is limited to textual claims and does not support multimodal verification (images, tables, charts). The rule-based claim decomposition, while fast and deterministic, may not capture all nuances of highly complex nested discourse structures. Future work will extend the framework to incorporate citation-graph neural networks for detecting multi-hop circular reporting chains, real-time streaming verification for conversational AI agents, and multimodal claim-evidence matching for vision-language model outputs.

6. References

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